KAIZEN MATHS

Lesson starter

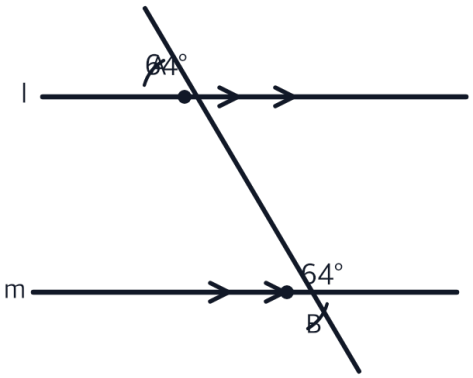
Formal Geometric Proof

Proof Foundations

Name _____

SA1

Name the theorem or angle fact that justifies each conclusion.



Lines l and m are parallel. The marked angles are equal. State the reason.

GEO-EB04-L1 · SA

Show your working

KAIZEN MATHS

Lesson starter

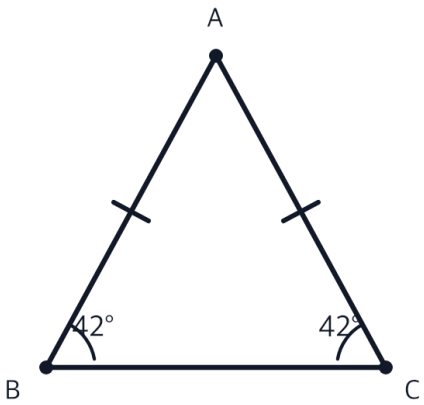
Formal Geometric Proof

Proof Foundations

Name _____

SB1

Use equal sides to identify equal base angles.



In triangle ABC, AB = AC. Explain why angle B equals angle C.

GEO-EB04-L1 · SB

Show your working

KAIZEN MATHS

Lesson starter

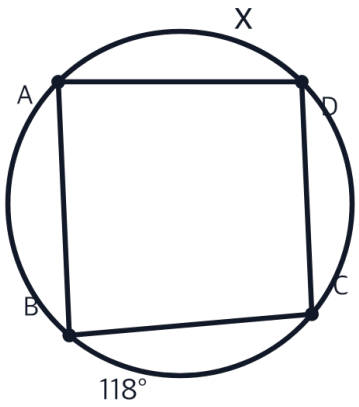
Formal Geometric Proof

Proof Foundations

Name _____

SC1

Use cyclic-quadrilateral angle facts to justify the conclusion.



ABCD is cyclic. Explain why the opposite angles add to 180°.

GEO-EB04-L1 · SC

Show your working

KAIZEN MATHS

Lesson starter

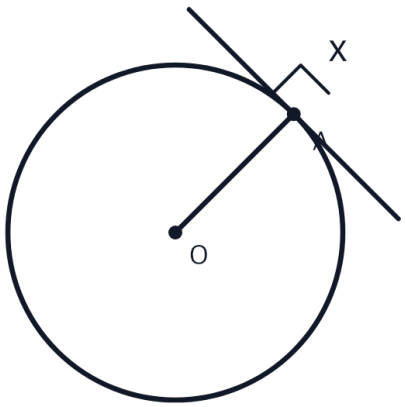
Formal Geometric Proof

Proof Foundations

Name _____

SD1

Use the tangent-radius theorem to justify the conclusion.



The line touches the circle at A. State the angle between the tangent and radius OA.

GEO-EB04-L1 · SD

Show your working

KAIZEN MATHS

Exit ticket

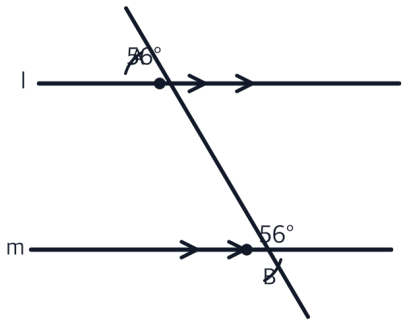
Formal Geometric Proof

Proof Foundations

Name _____

EA1

Name the theorem or angle fact that justifies each conclusion.



Lines l and m are parallel. The marked angles are equal. State the reason.

Show your working

KAIZEN MATHS

Exit ticket

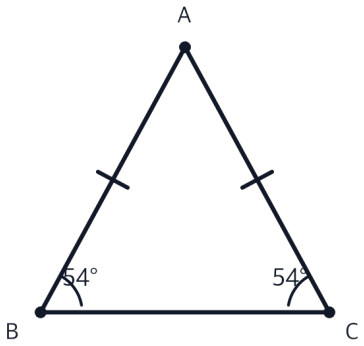
Formal Geometric Proof

Proof Foundations

Name _____

EB1

Use equal sides to identify equal base angles.



In triangle ABC, AB = AC. Explain why angle B equals angle C.

Show your working

KAIZEN MATHS

Exit ticket

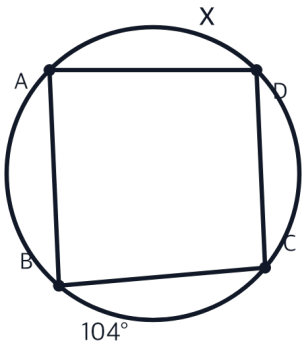
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Proof Foundations

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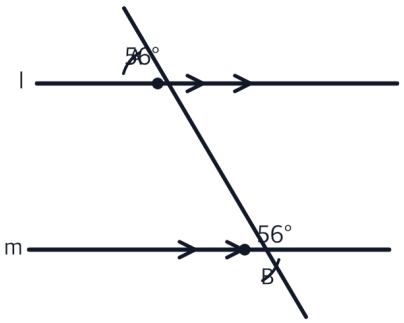
Formal Geometric Proof

Proof Foundations

Name _____

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Name the theorem or angle fact that justifies each conclusion.



Lines l and m are parallel. The marked angles are equal. State the reason.

Show your working

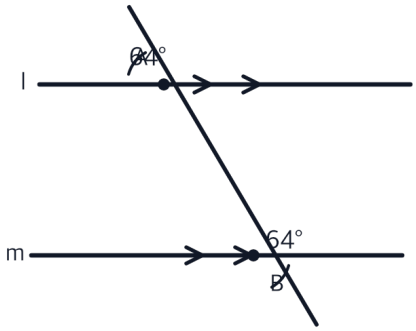
Teacher answers & working

Formal Geometric Proof · Proof Foundations

GEO-EB04-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Name the theorem or angle fact that justifies each conclusion.



Lines l and m are parallel. The marked angles are equal. State the reason.

Answer:

Alternate interior angles are equal.

Working:

The diagram shows l parallel to m.

The two marked angles are inside the parallel lines and on opposite sides of the transversal.

That relationship is called alternate interior angles.

Therefore the correct reason is: alternate interior angles are equal.

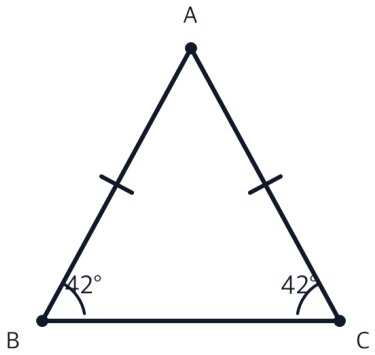
Teacher answers & working

Formal Geometric Proof · Proof Foundations

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SB1

Use equal sides to identify equal base angles.



In triangle ABC, AB = AC. Explain why angle B equals angle C.

Answer:

Base angles in an isosceles triangle are equal.

Working:

AB = AC, so triangle ABC has two equal sides.

The equal sides meet the base BC at B and C.

In an isosceles triangle, equal sides have equal opposite angles.

Therefore angle B = angle C.

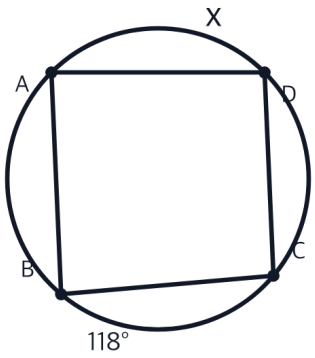
Teacher answers & working

Formal Geometric Proof · Proof Foundations

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SC1

Use cyclic-quadrilateral angle facts to justify the conclusion.



ABCD is cyclic. Explain why the opposite angles add to 180°.

Answer:

Opposite angles in a cyclic quadrilateral are supplementary.

Working:

All four vertices lie on the circumference, so ABCD is a cyclic quadrilateral.

A cyclic quadrilateral has opposite angles that are supplementary.

Supplementary means the two angles add to 180°.

Therefore angle B + angle D = 180°.

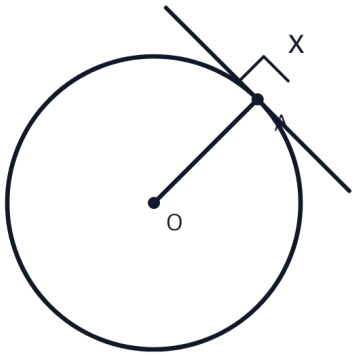
Teacher answers & working

Formal Geometric Proof · Proof Foundations

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SD1

Use the tangent-radius theorem to justify the conclusion.



The line touches the circle at A. State the angle between the tangent and radius OA.

Answer:

90°

Working:

The line through A is a tangent because it touches the circle at A.

OA is a radius drawn to the point of contact.

A tangent is perpendicular to the radius at the point of contact.

Therefore the angle between the tangent and OA is 90°.

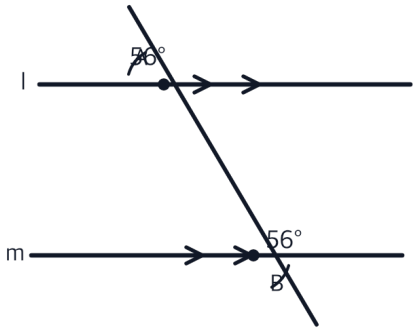
Teacher answers & working

Formal Geometric Proof · Proof Foundations

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EA1

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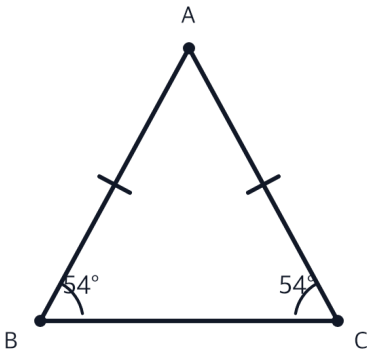
Teacher answers & working

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EB1

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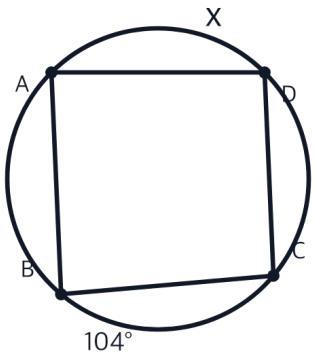
Teacher answers & working

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EC1

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